

MSTAR PRACTICE WORKSHEET**Lesson 5-6: Represent Three-Dimensional Figures in Two Dimensions**

Grade 6 Mathematics · Standard 6.GR.4

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 **SELECTED RESPONSE — CORRECT: C**

A rectangular prism has $l = 6$ in, $w = 4$ in, $h = 2$ in. What is its surface area?

- ☐ A 44 in²
- ☐ B 48 in²
- ☒ C 88 in²
- ☐ D 88 in³

$$SA = 2(6 \times 4) + 2(6 \times 2) + 2(4 \times 2) = 48 + 24 + 16 = 88 \text{ in}^2.$$

Question 2 **SELECTED RESPONSE — CORRECT: B**

How many faces does a rectangular prism have?

- ☐ A 4
- ☒ B 6
- ☐ C 8
- ☐ D 12

A rectangular prism has 6 faces: top, bottom, front, back, left, and right.

Question 3 **SELECTED RESPONSE — CORRECT: B**

Surface area is measured in which type of unit?

- ☐ A Cubic units (in^3 , cm^3 , ft^3)
- ☒ B Square units (in^2 , cm^2 , ft^2)
- ☐ C Linear units (in, cm, ft)
- ☐ D No units needed

Surface area measures flat surfaces, so it uses square units. Volume uses cubic units.

Question 4 **SELECTED RESPONSE — CORRECT: C**

A cube has edges of 5 cm. What is its surface area?

- ☐ A 30 cm^2
- ☐ B 125 cm^2
- ☒ C 150 cm^2
- ☐ D 150 cm^3

A cube has 6 identical faces. Each face: $5 \times 5 = 25 \text{ cm}^2$. Total SA = $6 \times 25 = 150 \text{ cm}^2$.

Question 5 **SELECTED RESPONSE — CORRECT: D**

You unfold a time capsule that is a cube. Its net shows 6 equal squares, and each square has an area of 4 cm^2 . What is the cube's total surface area?

- ☐ A 10 cm^2
- ☐ B 12 cm^2
- ☐ C 16 cm^2
- ☒ D 24 cm^2

A cube net has 6 equal square faces. With each face 4 cm^2 , the surface area is $6 \times 4 = 24 \text{ cm}^2$.

Question 6

SELECTED RESPONSE — CORRECT: C

Your team unfolds a rectangular time capsule box into a flat net. How many faces (flat sides) will the net show in all?

- ☐ A 4
- ☐ B 5
- ☒ C 6
- ☐ D 8

A rectangular prism has 6 faces — top, bottom, front, back, left, and right — so its net shows 6 faces.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7

TWO-PART QUESTION

Part A — correct answer: B

A net folds into a rectangular prism. How many rectangular faces does that net have?

- ☐ A 4
- ☒ B 6
- ☐ C 8
- ☐ D 12

A rectangular prism has 6 faces — top and bottom, front and back, and two ends — so its net shows all 6.

- **A:** That counts only the four side faces, leaving out the top and the bottom.
- **C:** A rectangular prism has 8 vertices, but the net shows its flat faces. Count those.
- **D:** A rectangular prism has 12 edges, but the net shows its flat faces. Count those.

Part B — correct answer: C

Which statement must be true for a flat pattern to be a valid net of a solid?

- ☐ A It must be a single long strip.
- ☐ B Every face must be the same size.
- ☒ C It folds into the solid with no faces overlapping and none missing.
- ☐ D It must contain only triangles.

A net is a flat pattern that folds into the solid with every face present exactly once and no overlap.

Question 8

SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about nets of three-dimensional figures:

- ☒ A A net shows every face of the solid laid flat.
- ☒ B A triangular prism net has 2 triangles and 3 rectangles.
- ☒ C The area of a net equals the surface area of the solid.
- ☐ D A net may have faces that overlap when folded.
- ☒ E One solid can have more than one correct net.

A, B, C, and E are true. D is false — overlapping faces mean the pattern does not fold into the solid, so it is not a net.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

Two boxes have the same volume of 60 in^3 . Box A is $5 \times 4 \times 3$. Box B is $10 \times 6 \times 1$. Which box uses less wrapping paper (has less surface area)? Show your calculations and explain why a more cube-like shape might use less material.

Model answer: Both boxes hold 60 in^3 , but wrapping paper is measured by surface area. Box A: $2(20) + 2(15) + 2(12) = 94 \text{ in}^2$. Box B: $2(60) + 2(10) + 2(6) = 152 \text{ in}^2$. Box A uses less paper because its dimensions are closer to a cube — flattening the same volume into $10 \times 6 \times 1$ creates two huge 60 in^2 faces and a lot of extra covering.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.